

SOGNEFJELL, Norway

WMO: 013660

Lat: 61.564N

Lon: 7.998E

Elev: 1413

StdP: 85.46

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-19.7	-25.6	0.4	-20.0	-23.4	0.5	-17.8	16.6	-7.0	15.2	-8.6	1.1	70	0.899

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	16.0	10.3	14.2	9.5	12.6	8.8	11.2	14.5	10.4	13.3	9.4	12.1	3.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.9	12.3	8.7	8.3	11.6	7.8	7.8	10.5	36.0	14.8	33.7	13.5	31.2	12.1	14.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
13.8	12.1	10.5		-24.0	17.1	3.0	1.6	-26.1	18.2	-27.9	19.2	-29.6	20.1	-31.7	21.3	
				-24.3	12.1	2.9	1.5	-26.4	13.2	-28.1	14.1	-29.8	15.0	-31.9	16.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-9.9	-9.2	-8.0	-4.0	0.3	4.4	7.8	7.0	3.5	-1.7	-6.1	-8.5
	DBStd	7.42	4.56	4.57	4.49	3.85	4.02	3.00	3.53	2.92	3.23	4.00	4.57	4.82
	HDD10.0	4408	617	537	559	419	302	170	87	101	196	362	483	573
	HDD18.3	7419	876	771	818	669	560	419	325	352	446	621	733	831
	CDD10.0	29	0	0	0	0	0	1	21	8	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.3	4.9	4.5	5.0	4.3	3.8	3.4	3.5	3.7	4.4	4.5	4.6	5.2
Precipitation	PrecAvg	796	88	68	65	36	37	49	63	68	70	80	82	91
	PrecMax	1101	231	160	281	64	92	114	110	108	120	194	165	190
	PrecMin	492	4	0	3	13	8	10	25	13	20	27	9	15
	PrecStd	131	56	44	54	14	18	26	22	24	29	38	45	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	1.7	3.3	7.4	13.5	15.1	19.4	16.8	13.5	9.6	4.7	2.0
		MCWB	-0.6	0.1	-0.5	2.8	7.6	8.0	12.1	10.1	8.7	7.1	2.0	1.9
	2%	DB	-1.7	-0.2	0.5	4.9	10.2	13.3	17.3	14.5	11.2	6.3	2.6	0.2
		MCWB	-2.0	-1.0	-0.9	1.5	5.9	7.5	11.3	9.8	7.9	4.2	1.4	0.0
	5%	DB	-2.8	-1.7	-0.8	3.4	8.0	11.0	15.5	12.7	9.7	4.9	1.2	-0.8
		MCWB	-3.2	-2.4	-1.6	1.0	4.5	6.5	10.4	9.4	7.4	3.2	0.3	-1.0
	10%	DB	-3.8	-3.1	-2.1	1.9	6.1	9.3	13.6	11.4	8.2	3.4	-0.4	-2.3
		MCWB	-4.2	-3.7	-2.9	-0.1	3.5	5.6	9.7	8.8	6.5	2.2	-0.9	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.8	1.0	3.7	8.2	9.2	12.9	11.4	10.3	7.7	3.2	1.9
		MCDB	-0.2	1.7	1.8	5.6	12.6	12.9	18.0	14.0	11.5	9.2	4.2	2.0
	2%	WB	-1.9	-0.8	-0.5	2.2	6.4	8.1	11.9	10.7	8.6	4.7	1.8	0.1
		MCDB	-1.8	-0.5	0.1	3.9	9.3	12.3	15.8	13.3	10.3	5.8	2.4	0.2
	5%	WB	-3.0	-2.3	-1.5	1.0	5.0	7.1	11.0	9.9	7.6	3.5	0.5	-0.9
		MCDB	-2.9	-1.9	-0.7	2.7	7.5	10.4	14.5	12.4	9.3	4.5	1.1	-0.8
	10%	WB	-4.1	-3.6	-2.8	-0.1	3.9	6.1	10.1	8.9	6.7	2.3	-0.9	-2.4
		MCDB	-3.9	-3.1	-2.1	1.5	5.9	8.7	13.3	11.3	8.2	3.2	-0.3	-2.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	6.1	6.5	5.8	5.9	5.3	4.4	4.2	4.4	4.9	5.2
		MCDBR	4.5	5.8	6.5	8.2	8.3	9.0	8.0	6.1	6.4	5.8	5.1	5.4
		MCWBR	4.5	5.5	5.4	6.0	5.0	5.4	4.0	3.2	4.0	4.6	4.5	5.5
	5% WB	MCDBR	4.3	5.4	5.7	6.8	7.4	8.0	7.1	5.4	5.4	5.2	4.9	5.5
		MCWBR	4.3	5.3	5.0	4.9	4.6	5.1	3.9	3.3	3.9	4.4	4.6	5.6
Clear-Sky Solar Irradiance	taub	0.182	0.225	0.254	0.282	0.289	0.291	0.301	0.294	0.268	0.246	0.188	0.150	
	taud	2.110	2.273	2.257	2.245	2.330	2.484	2.491	2.505	2.561	2.429	2.125	1.944	
	Ebn at Noon	594	792	879	914	935	939	919	900	872	760	565	441	
	Edh at Noon	33	59	87	109	109	96	93	84	65	50	32	22	
All-Sky Solar Radiation	RadAvg	0.24	0.83	2.17	3.95	4.92	4.97	4.64	3.57	2.24	1.03	0.33	0.14	
	RadStd	0.05	0.18	0.34	0.52	0.59	0.50	0.51	0.32	0.26	0.15	0.05	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+0.40	+0.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page